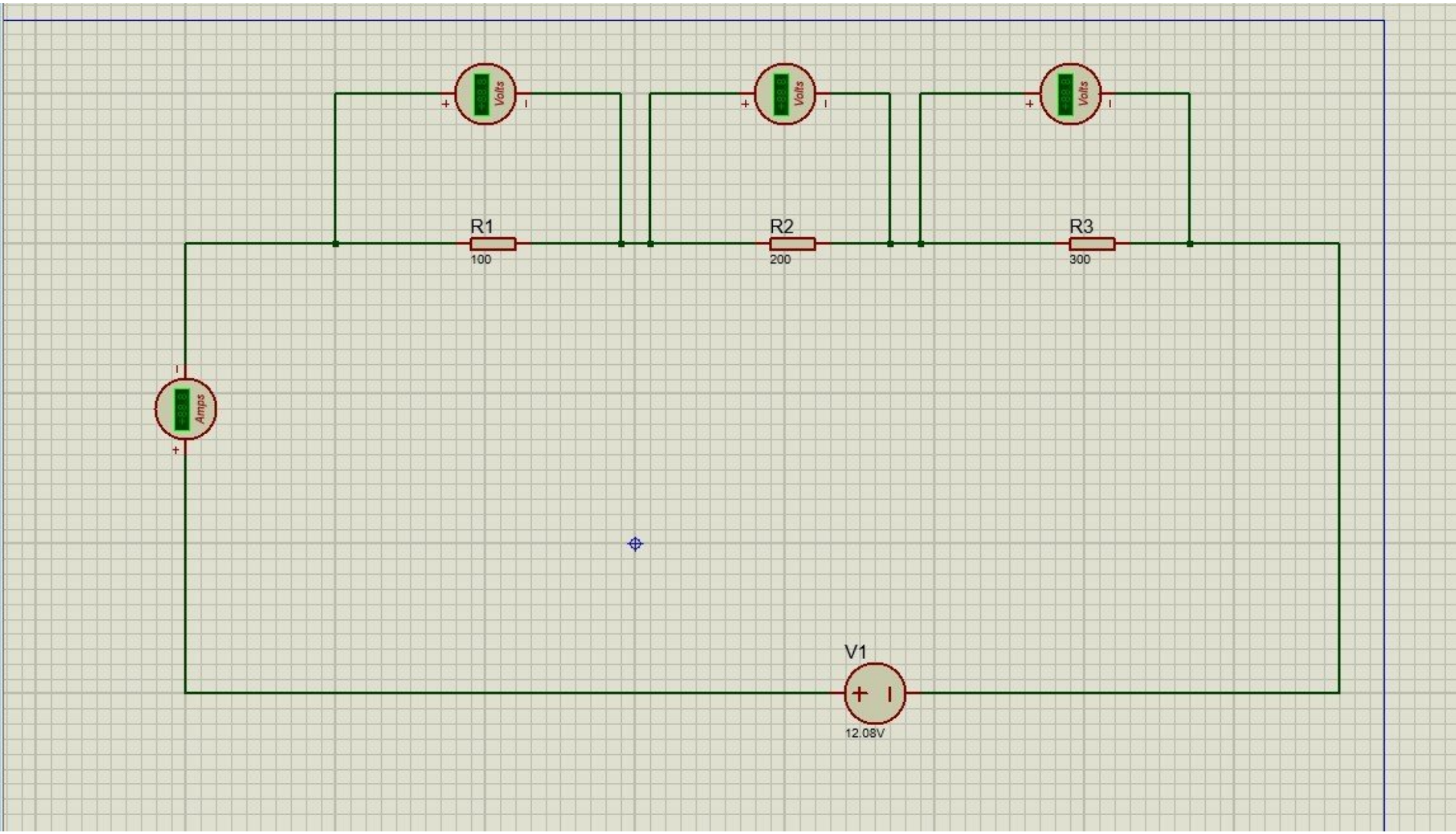
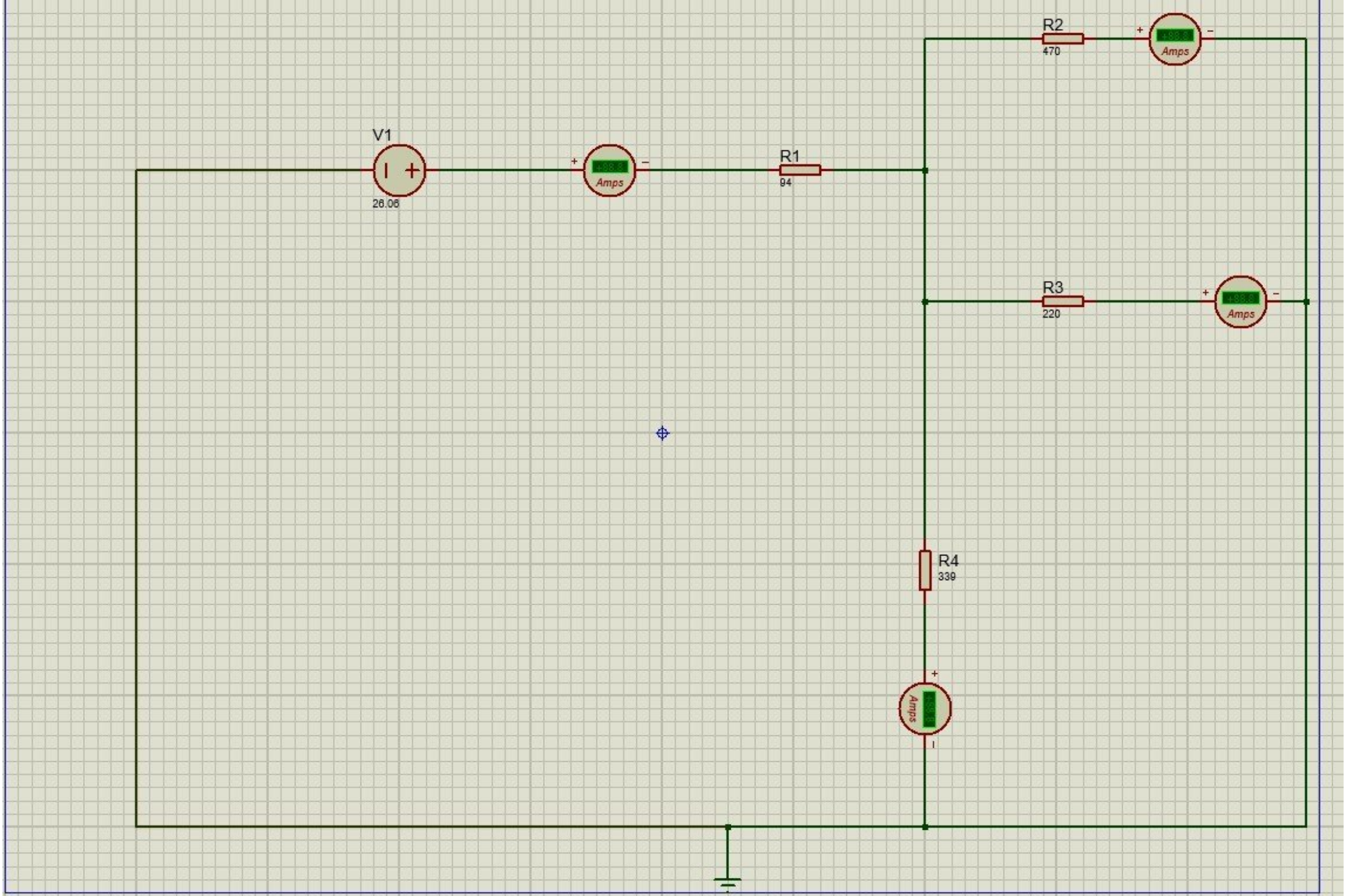

Volts
Volts
Volts
R1
100
R2
200
R3
300
Amps
V1
12.08V


R2
470
Amps
V1
26.06
Amps
R1
94
R3
220
Amps
R4
339
Amps

KVL

values (a)

$R_1 = 100\ \Omega$

$R_2 = 200\ \Omega$

$R_3 = 300\ \Omega$

$V = 12.08\ V$

i) calculation

$$R_T = R_1 + R_2 + R_3$$

$$R_T = 100 + 200 + 300$$

$$R_T = 600\ \Omega.$$

ii) calculation circuit current
ohm's law

$$I = \frac{V}{R_T}$$

$$I = \frac{12.08}{600}$$

$$I = 0.02013\ A$$

$$I = 0.02013 \times 1000$$

$$I = 20.13\ mA$$

iii) calculate voltage across R_1

$V_1 = I R_1$

$V_1 = 0.02013 \times 100$

$V_1 = 2.013V$

calculate voltage across R_2.

$V_2 = I R_2$

$V_2 = 0.02013 \times 200$

$V_2 = 4.027V$

calculate voltage across R_3

$V_3 = I R_3$

$V_3 = 0.02013 \times 300$

$V_3 = 6.040\ V$

KCl

values (b)

$R_1 = 94\ \Omega$

$R_2 = 470\ \Omega$

$R_3 = 220\ \Omega$

$R_4 = 339\ \Omega$

supply voltage } = 26.00V

i) calculate the equivalent resistance

first, R_2 and R_3 are connected in parallel

$$R_{23} = \frac{R_2 R_3}{R_2 + R_3}$$

$$R_{23} = \frac{470 \times 220}{470 + 220}$$

$$R_{23} = \frac{103400}{690}$$

$$R_{23} = 149.86\ \Omega.$$

R_{23} is in parallel with R_4

$$R_p = \frac{R_{23} R_4}{R_{23} + R_4}$$

$$= \frac{149.86 \times 339}{149.86 + 339}$$

$R_p = 103.93\ \Omega$

$R_T = R_1 + R_p$

$R_T = 94 + 103.93$

$R_T = 197.93\ \Omega.$

ii) calculate total current

$$I_T = \frac{V}{R_T}$$

$$I_T = \frac{26.06}{197.93}$$

$$I_T = 0.1317\ A$$

iii) calculate the current entering the junction.

$V_p = I_T \times R_p$

$V_p = 0.1317 \times 103.93$

$V_p = 13.69\ V.$

$$I_2 = \frac{V_p}{R_2} = \frac{13.69}{470}$$

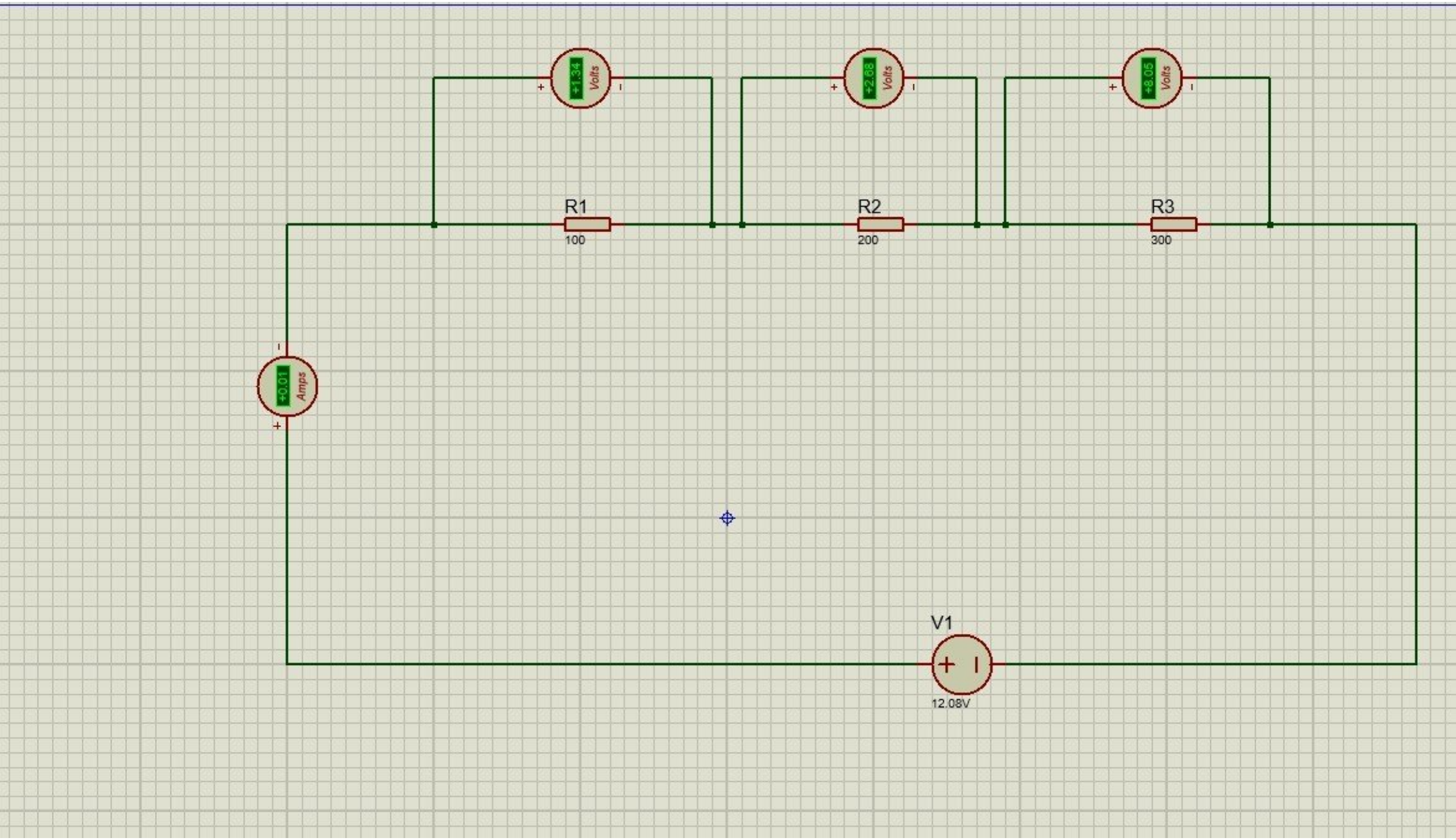
$$I_2 = 0.0291\ A$$

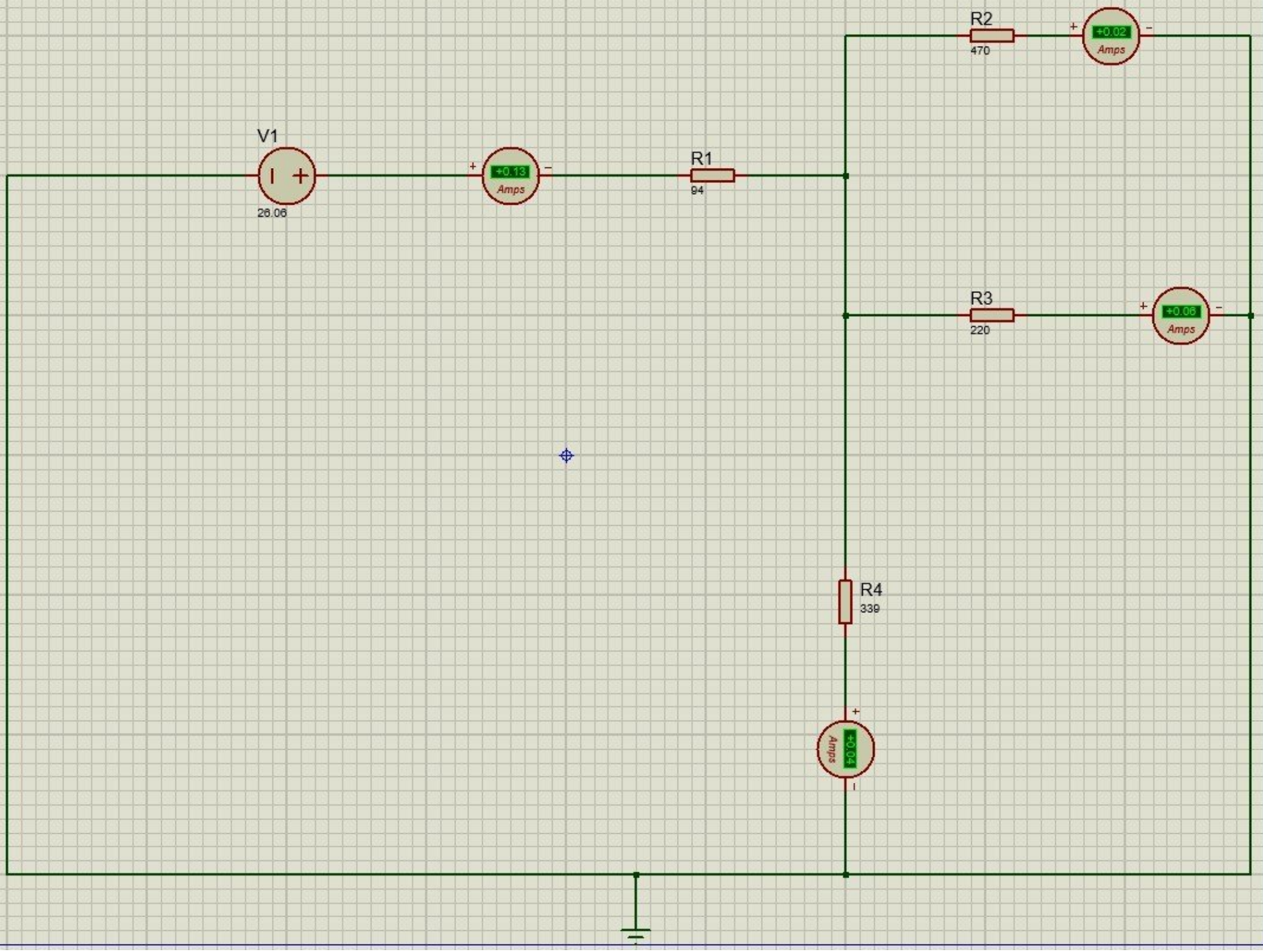
$$I_3 = \frac{V_p}{R_3} = \frac{13.69}{220}$$

$$I_3 = 0.0622\ A$$

$$I_4 = \frac{V_p}{R_4} = \frac{13.69}{339}$$

$$I_4 = 0.0404\ A$$


+1.34
Volts
+2.68
Volts
+8.05
Volts
R1
100
R2
200
R3
300
+0.01
Amps
V1
12.08V


R2
470
+0.02
Amps
V1
26.06
+0.13
Amps
R1
94
R3
220
+0.06
Amps
R4
339
+0.04
Amps

Tabulation

a) kvl

supply voltage(v)	V_1(v)	V_2(v)	V_3(v)	$V_1+V_2+V_3$(v)
12.08	2.013	4.027	6.040	12.08

$V_1+V_2+V_3 = 2.013+4.027+6.040$

$= 12.08V$

kvl is verified.

(b) Tabulation

I_1 (Practical) A	I_2 (A)	I_3 (A)	I_4 (A)	$I_2+I_3+I_4$
0.1317	0.0291	0.0622	0.0404	0.1317

$$I_2+I_3+I_4 = 0.0291+0.0622+0.0404$$
$$= 0.1317 \text{ A}$$

$$I_1 = 0.1317 \simeq 0.1317 \text{ A}$$

KCL is verified.